Healthcare Data Analysis

Sharon Kioni

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
healthcare_dataset = read.csv("healthcare_dataset.csv")
```

1. Print the Structure of the dataset

```
str(healthcare_dataset)
```

```
## 'data.frame':    55500 obs. of  15 variables:
##  $ Name              : chr  "Bobby JacksOn" "LesLie TErRy" "DaNnY sMitH" "andrEw waTtS" ...
##  $ Age               : int  30 62 76 28 43 36 21 20 82 58 ...
##  $ Gender            : chr  "Male" "Male" "Female" "Female" ...
##  $ Blood.Type        : chr  "B-" "A+" "A-" "O+" ...
##  $ Medical.Condition : chr  "Cancer" "Obesity" "Obesity" "Diabetes" ...
##  $ Date.of.Admission : chr  "2024-01-31" "2019-08-20" "2022-09-22" "2020-11-18" ...
##  $ Doctor            : chr  "Matthew Smith" "Samantha Davies" "Tiffany Mitchell" "Kevin Wells" ...
##  $ Hospital          : chr  "Sons and Miller" "Kim Inc" "Cook PLC" "Hernandez Rogers and Vang," ...
##  $ Insurance.Provider: chr  "Blue Cross" "Medicare" "Aetna" "Medicare" ...
##  $ Billing.Amount    : num  18856 33643 27955 37910 14238 ...
##  $ Room.Number       : int  328 265 205 450 458 389 389 277 316 249 ...
##  $ Admission.Type    : chr  "Urgent" "Emergency" "Emergency" "Elective" ...
##  $ Discharge.Date    : chr  "2024-02-02" "2019-08-26" "2022-10-07" "2020-12-18" ...
##  $ Medication        : chr  "Paracetamol" "Ibuprofen" "Aspirin" "Ibuprofen" ...
##  $ Test.Results      : chr  "Normal" "Inconclusive" "Normal" "Abnormal" ...
```

2. List the variables in the dataset

```
names(healthcare_dataset)
```

```
##  [1] "Name"               "Age"                "Gender"
##  [4] "Blood.Type"         "Medical.Condition"  "Date.of.Admission"
##  [7] "Doctor"             "Hospital"           "Insurance.Provider"
## [10] "Billing.Amount"     "Room.Number"        "Admission.Type"
## [13] "Discharge.Date"     "Medication"         "Test.Results"
```

3. Print the top 15 rows of the dataset

```
head(healthcare_dataset,15)
```

```
##                 Name Age Gender Blood.Type Medical.Condition Date.of.Admission
## 1       Bobby JacksOn  30   Male         B-            Cancer        2024-01-31
## 2        LesLie TErRy  62   Male         A+           Obesity        2019-08-20
## 3         DaNnY sMitH  76 Female         A-           Obesity        2022-09-22
## 4        andrEw waTtS  28 Female         O+          Diabetes        2020-11-18
## 5       adrIENNE bEll  43 Female        AB+            Cancer        2022-09-19
## 6       EMILY JOHNSOn  36   Male         A+            Asthma        2023-12-20
## 7      edwArD EDWaRDs  21 Female        AB-          Diabetes        2020-11-03
## 8  CHrisTInA MARtinez  20 Female         A+            Cancer        2021-12-28
## 9     JASmINe aGuIlaR  82   Male        AB+            Asthma        2020-07-01
## 10   ChRISTopher BerG  58 Female        AB-            Cancer        2021-05-23
## 11   mIchElLe daniELs  72   Male         O+            Cancer        2020-04-19
## 12     aaRon MARtiNeZ  38 Female         A-      Hypertension        2023-08-13
## 13      connOR HANsEn  75 Female         A+          Diabetes        2019-12-12
## 14       rObeRt bAuer  68 Female        AB+            Asthma        2020-05-22
## 15       bROOkE brady  44 Female        AB+            Cancer        2021-10-08
##              Doctor                     Hospital Insurance.Provider
## 1     Matthew Smith            Sons and Miller         Blue Cross
## 2   Samantha Davies                   Kim Inc           Medicare
## 3  Tiffany Mitchell                  Cook PLC              Aetna
## 4       Kevin Wells  Hernandez Rogers and Vang,           Medicare
## 5    Kathleen Hanna               White-White              Aetna
## 6     Taylor Newton             Nunez-Humphrey   UnitedHealthcare
## 7       Kelly Olson            Group Middleton           Medicare
## 8    Suzanne Thomas Powell Robinson and Valdez,              Cigna
## 9   Daniel Ferguson              Sons Rich and              Cigna
## 10      Heather Day             Padilla-Walker   UnitedHealthcare
## 11      John Duncan            Schaefer-Porter           Medicare
## 12     Douglas Mayo                Lyons-Blair           Medicare
## 13 Kenneth Fletcher   Powers Miller, and Flores              Cigna
## 14  Theresa Freeman           Rivera-Gutierrez   UnitedHealthcare
## 15  Roberta Stewart            Morris-Arellano   UnitedHealthcare
##    Billing.Amount Room.Number Admission.Type Discharge.Date  Medication
## 1       18856.281         328         Urgent     2024-02-02 Paracetamol
## 2       33643.327         265      Emergency     2019-08-26   Ibuprofen
## 3       27955.096         205      Emergency     2022-10-07     Aspirin
```

```
## 4       37909.782         450       Elective     2020-12-18   Ibuprofen
## 5       14238.318         458         Urgent     2022-10-09  Penicillin
## 6       48145.111         389         Urgent     2023-12-24   Ibuprofen
## 7       19580.872         389      Emergency     2020-11-15 Paracetamol
## 8       45820.463         277      Emergency     2022-01-07 Paracetamol
## 9       50119.223         316       Elective     2020-07-14     Aspirin
## 10      19784.631         249       Elective     2021-06-22 Paracetamol
## 11      12576.796         394         Urgent     2020-04-22 Paracetamol
## 12       7999.587         288         Urgent     2023-09-05     Lipitor
## 13      43282.283         134      Emergency     2019-12-28  Penicillin
## 14      33207.707         309         Urgent     2020-06-19     Lipitor
## 15      40701.599         182         Urgent     2021-10-13 Paracetamol
##    Test.Results
## 1        Normal
## 2  Inconclusive
## 3        Normal
## 4      Abnormal
## 5      Abnormal
## 6        Normal
## 7  Inconclusive
## 8  Inconclusive
## 9      Abnormal
## 10 Inconclusive
## 11       Normal
## 12 Inconclusive
## 13     Abnormal
## 14       Normal
## 15       Normal
```

4. Write a user defined function using a variable in the dataset

```
average_age = function(age)(mean(age, na.rm = TRUE))
average_age(healthcare_dataset$Age)
```

```
## [1] 51.53946
```

5. Use data manipulation techniques and filter rows based on any logical criteria that exist in the dataset

```
filtered_healthcare = subset(healthcare_dataset, Age >60 & Billing.Amount >30000,select = c(Name, Age, M
head(filtered_healthcare,15)
```

```
##                Name Age Medical.Condition Billing.Amount
## 2      LesLie TErRy  62           Obesity       33643.33
## 9   JASmINe aGuIlaR  82            Asthma       50119.22
## 13    connOR HANsEn  75          Diabetes       43282.28
## 14     rObeRt bAuer  68            Asthma       33207.71
## 25   PAUl hEndERsOn  72      Hypertension       33211.30
## 39    NIcOlE LUcErO  70          Diabetes       48290.69
```

```
## 41  chRiSTOPHer LEe  74      Hypertension       49943.28
## 42     ThomAS morsE  81      Hypertension       35633.96
## 45    pAUL wILLiAmS  81            Cancer       42684.56
## 46    lYNn MaRtinez  65            Cancer       45585.89
## 62        Kim ScOtt  63            Asthma       39723.17
## 82      ChAd MorEnO  67      Hypertension       46814.01
## 87    kayLa padillA  63            Asthma       34774.35
## 105    HunTeR mckAy  85         Arthritis       40014.76
## 123  Jeffrey tuRNER  85           Obesity       39957.94
```

6. Identify the dependent and independent variables and use reshaping techniques and create a new data frame by joining those variables from your dataset

```
independent = healthcare_dataset[, c("Name", "Age", "Medical.Condition")]
dependent = healthcare_dataset[, c("Name", "Age", "Billing.Amount")]
joined_healthcare = merge(independent, dependent, by ="Name")
head(joined_healthcare,15)
```

```
##                 Name Age.x Medical.Condition Age.y Billing.Amount
## 1        aAroN ADaMS    38            Cancer    38      26052.106
## 2      aAROn aguIRRe    36          Diabetes    36      27087.561
## 3     AArOn AnderSoN    50            Asthma    50      39804.659
## 4  aaRON AndeRSoN md    20      Hypertension    20      16846.416
## 5       AAron ArCHER    49            Cancer    49      10602.077
## 6       AAron ArCHER    49            Cancer    47      10602.077
## 7       AAron ArCHER    47            Cancer    49      10602.077
## 8       AAron ArCHER    47            Cancer    47      10602.077
## 9        AaROn bAkEr    84            Asthma    84       6826.677
## 10       AAroN bAker    73            Cancer    73      10135.885
## 11 AARON bAldWIN Jr.    20      Hypertension    20      29740.960
## 12      aaROn bARNes    85           Obesity    85       4783.444
## 13    aaROn bauTISta    21      Hypertension    21      49827.520
## 14     aarOn bEnNetT    76          Diabetes    76      12740.887
## 15     AArON BeNtlEy    18            Asthma    18       5411.611
```

7. Remove missing values in the dataset

```
sum(is.na(healthcare_dataset))
```

```
## [1] 0
```

#Clean healthcare dateset

```
clean_healthcare = na.omit(healthcare_dataset)
```

8. Identify and remove duplicated data in the dataset

```r
sum(duplicated(healthcare_dataset))
```

```
## [1] 534
```

#Remove duplicate data

```r
healthcare_unique = unique(healthcare_dataset)
```

#check if the duplicates have been removed

```r
sum(duplicated(healthcare_unique))
```

```
## [1] 0
```

9. Reorder multiple rows in descending order

```r
sorted_healthcare = healthcare_unique[order(-healthcare_unique$Age,-healthcare_unique$Billing.Amount),]
head(sorted_healthcare,15)
```

```
##                       Name Age Gender Blood.Type Medical.Condition
## 52858      THomAs PHIlllpS  89 Female         A+      Hypertension
## 52044         DAVId NeWTOn  89 Female         O+         Arthritis
## 55386       hEAthEr dawsOn  89   Male         O-         Arthritis
## 52829         doNALD aViLA  89 Female        AB-            Asthma
## 53826        mIchaeL POtTs  89   Male         A+      Hypertension
## 54045    MiChAEL DOmINGuEz  89   Male         O+            Cancer
## 54814    JerEmY hArdIN JR.  89   Male         A+          Diabetes
## 52373      DeBORah McBrIdE  89 Female         O+           Obesity
## 53597          PeTEr bAuer  88 Female         O+      Hypertension
## 51639        WIlliaM cAREY  88   Male        AB-            Cancer
## 52943 mr. fRANKLIN cOllInS  88 Female         A+      Hypertension
## 52845          DAIsY YouNg  88   Male         B-      Hypertension
## 53642    FelIcIa VELAzqUeZ  88   Male         B-          Diabetes
## 50322       BARBaRA OSbOrN  88 Female         O+            Asthma
## 50003      DaNiEL THoMPSon  88 Female        AB+      Hypertension
##       Date.of.Admission            Doctor                     Hospital
## 52858        2022-05-22        Mark White    and Martin, Davidson Cox
## 52044        2021-02-08     Jerry Hopkins     Cooper Brown Parks, and
## 55386        2022-07-18          Ann Hall                  Moore-Gray
## 52829        2022-09-17 Christopher Allen   Holmes and Howard Castro,
## 53826        2023-01-19       Mary Vaughn and Perez, Bennett Townsend
## 54045        2021-09-07     Bridget Irwin                   PLC White
## 54814        2019-11-05    Willie Stevens                Gray-Solomon
## 52373        2021-05-28     Robert Garcia     and Miller, Olson Olson
## 53597        2019-07-28       Jason Pratt                  Harris LLC
## 51639        2020-03-14      Joshua Pitts             Sons and Prince
## 52943        2023-04-07   Alyssa Lawrence                 LLC Hancock
## 52845        2023-03-12        Beth Adams             Brooks-Martinez
## 53642        2023-05-20         Angela Le     Rogers, Wiley and Ortiz
```

```
## 50322        2020-03-10     Robert Medina             Rangel-Phillips
## 50003        2023-12-14        Ryan Wolfe                Davis-Hicks
##       Insurance.Provider Billing.Amount Room.Number Admission.Type
## 52858              Cigna      48274.349         332         Urgent
## 52044              Aetna      34500.017         242       Elective
## 55386              Aetna      17172.116         455       Elective
## 52829              Aetna      14042.749         199      Emergency
## 53826              Cigna       9542.740         289      Emergency
## 54045              Aetna       7628.951         363       Elective
## 54814              Cigna       7242.641         113       Elective
## 52373              Cigna       5692.907         166       Elective
## 53597              Aetna      47363.627         409         Urgent
## 51639         Blue Cross      45057.769         163         Urgent
## 52943         Blue Cross      44655.537         321       Elective
## 52845              Aetna      42288.179         420      Emergency
## 53642         Blue Cross      40720.450         114         Urgent
## 50322           Medicare      31051.589         476      Emergency
## 50003         Blue Cross      29177.755         157       Elective
##       Discharge.Date  Medication Test.Results
## 52858     2022-05-26  Penicillin Inconclusive
## 52044     2021-02-20  Penicillin Inconclusive
## 55386     2022-08-13   Ibuprofen       Normal
## 52829     2022-09-27     Aspirin     Abnormal
## 53826     2023-01-28 Paracetamol     Abnormal
## 54045     2021-09-24     Aspirin     Abnormal
## 54814     2019-11-28  Penicillin       Normal
## 52373     2021-06-13 Paracetamol     Abnormal
## 53597     2019-08-18     Lipitor Inconclusive
## 51639     2020-04-07     Lipitor       Normal
## 52943     2023-05-04     Lipitor Inconclusive
## 52845     2023-03-22     Aspirin Inconclusive
## 53642     2023-05-23     Aspirin     Abnormal
## 50322     2020-03-29   Ibuprofen       Normal
## 50003     2023-12-16     Lipitor       Normal
```

10. Rename some of the column names in the dataset

```
colnames(healthcare_dataset)
```

```
##  [1] "Name"               "Age"                "Gender"
##  [4] "Blood.Type"         "Medical.Condition"  "Date.of.Admission"
##  [7] "Doctor"             "Hospital"           "Insurance.Provider"
## [10] "Billing.Amount"     "Room.Number"        "Admission.Type"
## [13] "Discharge.Date"     "Medication"         "Test.Results"
```

#Rename

```
renamed_healthcare = sorted_healthcare
names(renamed_healthcare)[2] = "Patient_Age"
names(renamed_healthcare)[5] = "Diagnosis"
names(renamed_healthcare)[10] = "Medical_Bill"
head(renamed_healthcare,15)
```

```
##                       Name Patient_Age Gender Blood.Type    Diagnosis
## 52858      THomAs PHIlllpS          89 Female         A+ Hypertension
## 52044         DAVId NeWTOn          89 Female         O+    Arthritis
## 55386       hEAthEr dawsOn          89   Male         O-    Arthritis
## 52829         doNALD aViLA          89 Female        AB-       Asthma
## 53826        mIchaeL POtTs          89   Male         A+ Hypertension
## 54045    MiChAEL DOmINGuEz          89   Male         O+       Cancer
## 54814    JerEmY hArdIN JR.          89   Male         A+     Diabetes
## 52373      DeBORah McBrIdE          89 Female         O+      Obesity
## 53597           PeTEr bAuer          88 Female         O+ Hypertension
## 51639        WIlliaM cAREY          88   Male        AB-       Cancer
## 52943 mr. fRANKLIN cOllInS          88 Female         A+ Hypertension
## 52845          DAIsY YouNg          88   Male         B- Hypertension
## 53642    FelIcIa VELAzqUeZ          88   Male         B-     Diabetes
## 50322       BARBaRA OSbOrN          88 Female         O+       Asthma
## 50003      DaNiEL THoMPSon          88 Female        AB+ Hypertension
##       Date.of.Admission            Doctor                      Hospital
## 52858        2022-05-22        Mark White    and Martin, Davidson Cox
## 52044        2021-02-08     Jerry Hopkins     Cooper Brown Parks, and
## 55386        2022-07-18          Ann Hall                 Moore-Gray
## 52829        2022-09-17 Christopher Allen   Holmes and Howard Castro,
## 53826        2023-01-19       Mary Vaughn and Perez, Bennett Townsend
## 54045        2021-09-07     Bridget Irwin                   PLC White
## 54814        2019-11-05    Willie Stevens                Gray-Solomon
## 52373        2021-05-28     Robert Garcia     and Miller, Olson Olson
## 53597        2019-07-28       Jason Pratt                 Harris LLC
## 51639        2020-03-14      Joshua Pitts             Sons and Prince
## 52943        2023-04-07   Alyssa Lawrence                 LLC Hancock
## 52845        2023-03-12        Beth Adams             Brooks-Martinez
## 53642        2023-05-20         Angela Le     Rogers, Wiley and Ortiz
## 50322        2020-03-10     Robert Medina             Rangel-Phillips
## 50003        2023-12-14        Ryan Wolfe                 Davis-Hicks
##       Insurance.Provider Medical_Bill Room.Number Admission.Type Discharge.Date
## 52858              Cigna    48274.349         332         Urgent     2022-05-26
## 52044              Aetna    34500.017         242       Elective     2021-02-20
## 55386              Aetna    17172.116         455       Elective     2022-08-13
## 52829              Aetna    14042.749         199      Emergency     2022-09-27
## 53826              Cigna     9542.740         289      Emergency     2023-01-28
## 54045              Aetna     7628.951         363       Elective     2021-09-24
## 54814              Cigna     7242.641         113       Elective     2019-11-28
## 52373              Cigna     5692.907         166       Elective     2021-06-13
## 53597              Aetna    47363.627         409         Urgent     2019-08-18
## 51639        Blue Cross    45057.769         163         Urgent     2020-04-07
## 52943        Blue Cross    44655.537         321       Elective     2023-05-04
## 52845              Aetna    42288.179         420      Emergency     2023-03-22
## 53642        Blue Cross    40720.450         114         Urgent     2023-05-23
## 50322          Medicare    31051.589         476      Emergency     2020-03-29
## 50003        Blue Cross    29177.755         157       Elective     2023-12-16
##        Medication Test.Results
## 52858  Penicillin Inconclusive
## 52044  Penicillin Inconclusive
## 55386   Ibuprofen       Normal
## 52829     Aspirin     Abnormal
## 53826 Paracetamol     Abnormal
```

```
## 54045     Aspirin     Abnormal
## 54814  Penicillin       Normal
## 52373 Paracetamol     Abnormal
## 53597     Lipitor Inconclusive
## 51639     Lipitor       Normal
## 52943     Lipitor Inconclusive
## 52845     Aspirin Inconclusive
## 53642     Aspirin     Abnormal
## 50322   Ibuprofen       Normal
## 50003     Lipitor       Normal
```

11. Add new variables in your data frame by using a mathematical function (for e.g. – multiply an existing column by 2 and add it as a new variable to your data frame

```
renamed_healthcare$Double_Bill = renamed_healthcare$Medical_Bill * 2
head(renamed_healthcare[, c("Medical_Bill", "Double_Bill")], 15)
```

```
##       Medical_Bill Double_Bill
## 52858    48274.349    96548.70
## 52044    34500.017    69000.03
## 55386    17172.116    34344.23
## 52829    14042.749    28085.50
## 53826     9542.740    19085.48
## 54045     7628.951    15257.90
## 54814     7242.641    14485.28
## 52373     5692.907    11385.81
## 53597    47363.627    94727.25
## 51639    45057.769    90115.54
## 52943    44655.537    89311.07
## 52845    42288.179    84576.36
## 53642    40720.450    81440.90
## 50322    31051.589    62103.18
## 50003    29177.755    58355.51
```

12. Create a training set using random number generator engine.

```
library(dplyr)
set.seed(1234)
Training_healthcare = renamed_healthcare %>% sample_frac(0.80, rep = FALSE)
Training_healthcare = as.data.frame(Training_healthcare)
head(Training_healthcare, 15)
```

```
##                  Name Patient_Age Gender Blood.Type    Diagnosis
## 1         MarY dAniEL          35   Male        AB+       Asthma
## 2     KaThleeN Taylor          35   Male         A- Hypertension
## 3      NIchOLAs cAsey          34 Female        AB+ Hypertension
## 4    ELizaBETH PoRter          67   Male         B- Hypertension
## 5        daNIEL PatEl          44   Male         O+ Hypertension
```

```
## 6        JaMES TOrREs          41   Male         A+    Arthritis
## 7        mARIA gaRNer          17   Male         O- Hypertension
## 8  CHRIstOPheR hARRIS          64 Female         A-    Arthritis
## 9      ShanNon HOrtoN          67 Female        AB+    Arthritis
## 10      daniEl poTtEr          61   Male         B+      Obesity
## 11     miChAeL MorGan          82   Male         O+      Obesity
## 12    JeSsIca perkinS          20   Male         B-       Asthma
## 13  sCotT CHrisTenseN          82   Male         O+       Cancer
## 14        bONNIE crUZ          64 Female         O-    Arthritis
## 15        JamES bRoWn          44 Female         A-     Diabetes
##    Date.of.Admission            Doctor                     Hospital
## 1         2021-12-12 Christine Padilla                  Gill-Munoz
## 2         2022-06-01     Jennifer King              and Munoz Sons
## 3         2021-07-26       Rachel Peck               Allen-Johnson
## 4         2022-09-26      Olivia Moore             Fuller and Sons
## 5         2020-12-10       Russell Fox               Roberts Group
## 6         2020-12-14     Jessica Pratt    Stone, Ingram and Deleon
## 7         2019-09-26     Robin Johnson  Bryant, Hendrix and Rangel
## 8         2020-01-15     Ricky Marquez             Ibarra-Campbell
## 9         2022-07-01     Jared Mendoza Johnson, Edwards Miller and
## 10        2019-10-25       Chloe Kelly    Brown and Vincent Roman,
## 11        2020-01-08        Ian Snyder     and Rivera Mccoy Foley,
## 12        2022-03-27      Richard Ryan    Perez and Green, Carlson
## 13        2020-02-21   Melissa Mendoza                 Hill-Perez
## 14        2022-04-13   Catherine Adams            Anderson-Jimenez
## 15        2022-01-01   Samantha Jordan                LLC Bradley
##    Insurance.Provider Medical_Bill Room.Number Admission.Type Discharge.Date
## 1          Blue Cross    36766.177         140      Emergency     2022-01-09
## 2            Medicare    32427.846         354      Emergency     2022-06-22
## 3               Aetna    15687.441         334         Urgent     2021-08-25
## 4          Blue Cross     1598.057         135      Emergency     2022-10-23
## 5               Cigna    22164.607         386       Elective     2020-12-18
## 6    UnitedHealthcare    50405.041         370       Elective     2020-12-18
## 7    UnitedHealthcare    17889.765         293      Emergency     2019-10-07
## 8               Aetna    17289.893         439       Elective     2020-01-20
## 9            Medicare     2500.236         187         Urgent     2022-07-23
## 10           Medicare    22012.015         395         Urgent     2019-11-15
## 11         Blue Cross    34642.792         193      Emergency     2020-01-16
## 12              Cigna    11615.585         342      Emergency     2022-04-01
## 13           Medicare    11515.498         304         Urgent     2020-03-20
## 14   UnitedHealthcare    23265.039         380         Urgent     2022-04-18
## 15              Cigna    48859.294         136       Elective     2022-01-23
##    Medication Test.Results Double_Bill
## 1     Lipitor       Normal   73532.354
## 2     Lipitor     Abnormal   64855.692
## 3     Lipitor       Normal   31374.881
## 4     Lipitor Inconclusive    3196.114
## 5  Penicillin       Normal   44329.213
## 6  Penicillin     Abnormal  100810.081
## 7     Aspirin Inconclusive   35779.530
## 8     Lipitor     Abnormal   34579.785
## 9     Lipitor Inconclusive    5000.471
## 10  Ibuprofen Inconclusive   44024.029
## 11  Ibuprofen     Abnormal   69285.584
```

```
## 12  Ibuprofen Inconclusive   23231.169
## 13 Penicillin Inconclusive   23030.996
## 14    Lipitor     Abnormal   46530.078
## 15    Lipitor Inconclusive   97718.588
```

12. Print the summary statistics of the dataset.

```
summary(Training_healthcare)
```

```
##         Name        Patient_Age         Gender          Blood.Type
##  Length   :43973   Min.   :13.0   Length   :43973   Length   :43973
##  N.unique :40811   1st Qu.:35.0   N.unique :    2   N.unique :    8
##  N.blank  :    0   Median :52.0   N.blank  :    0   N.blank  :    0
##  Min.nchar:    6   Mean   :51.5   Min.nchar:    4   Min.nchar:    2
##  Max.nchar:   28   3rd Qu.:68.0   Max.nchar:    6   Max.nchar:    3
##                    Max.   :89.0
##      Diagnosis     Date.of.Admission       Doctor           Hospital
##  Length   :43973   Length   :43973   Length   :43973   Length   :43973
##  N.unique :    6   N.unique : 1827   N.unique :33855   N.unique :33242
##  N.blank  :    0   N.blank  :    0   N.blank  :    0   N.blank  :    0
##  Min.nchar:    6   Min.nchar:   10   Min.nchar:    6   Min.nchar:    6
##  Max.nchar:   12   Max.nchar:   10   Max.nchar:   28   Max.nchar:   35
##
##  Insurance.Provider  Medical_Bill    Room.Number      Admission.Type
##  Length   :43973    Min.   :-2008   Min.   :101.0   Length   :43973
##  N.unique :    5    1st Qu.:13236   1st Qu.:202.0   N.unique :    3
##  N.blank  :    0    Median :25545   Median :302.0   N.blank  :    0
##  Min.nchar:    5    Mean   :25556   Mean   :300.9   Min.nchar:    6
##  Max.nchar:   16    3rd Qu.:37855   3rd Qu.:400.0   Max.nchar:    9
##                     Max.   :52764   Max.   :500.0
##    Discharge.Date      Medication       Test.Results    Double_Bill
##  Length   :43973   Length   :43973   Length   :43973   Min.   : -4017
##  N.unique : 1855   N.unique :    5   N.unique :    3   1st Qu.: 26473
##  N.blank  :    0   N.blank  :    0   N.blank  :    0   Median : 51090
##  Min.nchar:   10   Min.nchar:    7   Min.nchar:    6   Mean   : 51113
##  Max.nchar:   10   Max.nchar:   11   Max.nchar:   12   3rd Qu.: 75710
##                                                       Max.   :105529
```

13. Use any of the numerical variables from the dataset and perform the following statistical functions(Mean, Median, Mode,Range.

#mean

```
mean(Training_healthcare$Patient_Age)
```

```
## [1] 51.49599
```

#median

```
median(Training_healthcare$Patient_Age)
```

```
## [1] 52
```

#mode

```
Age_frequency = table(Training_healthcare$Patient_Age)
names(Age_frequency)[which.max(Age_frequency)]
```

```
## [1] "38"
```

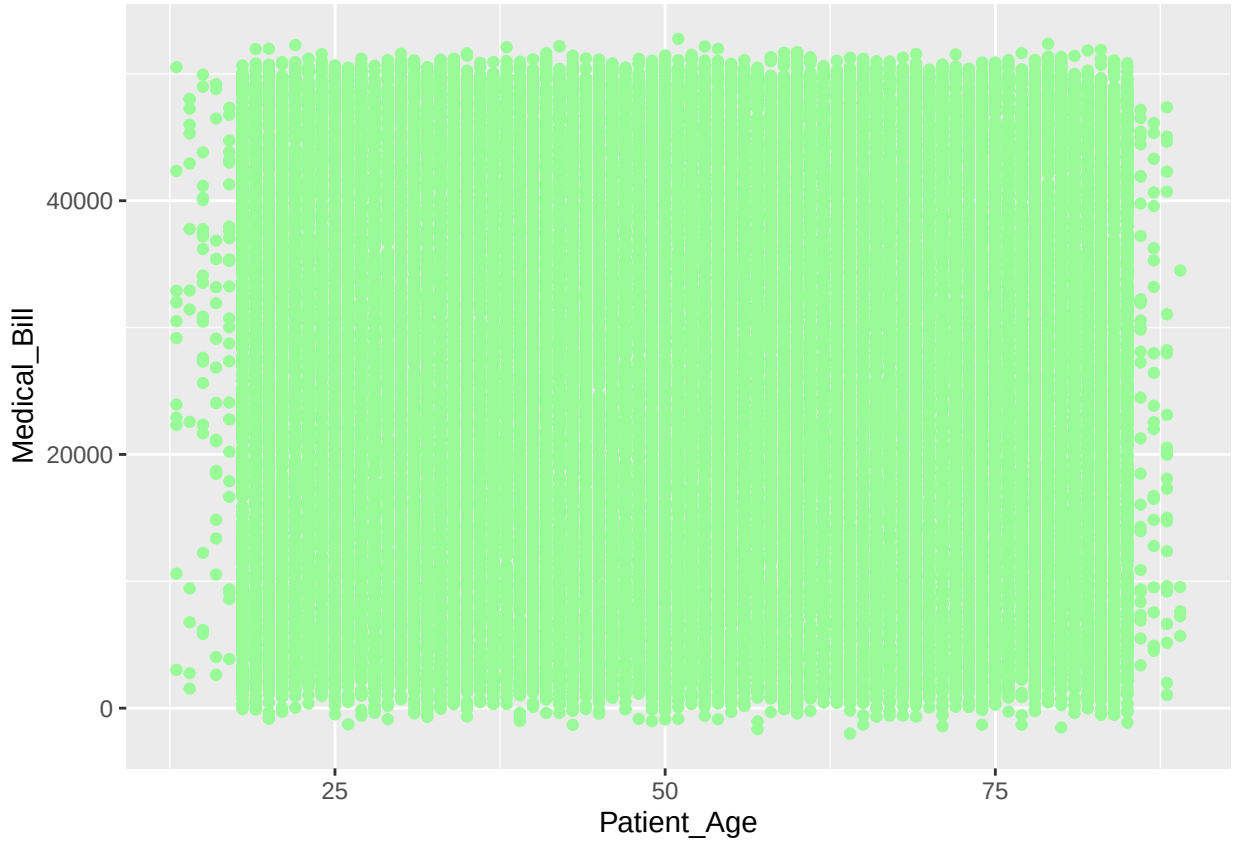
#Range

```
range(Training_healthcare$Patient_Age)
```

```
## [1] 13 89
```

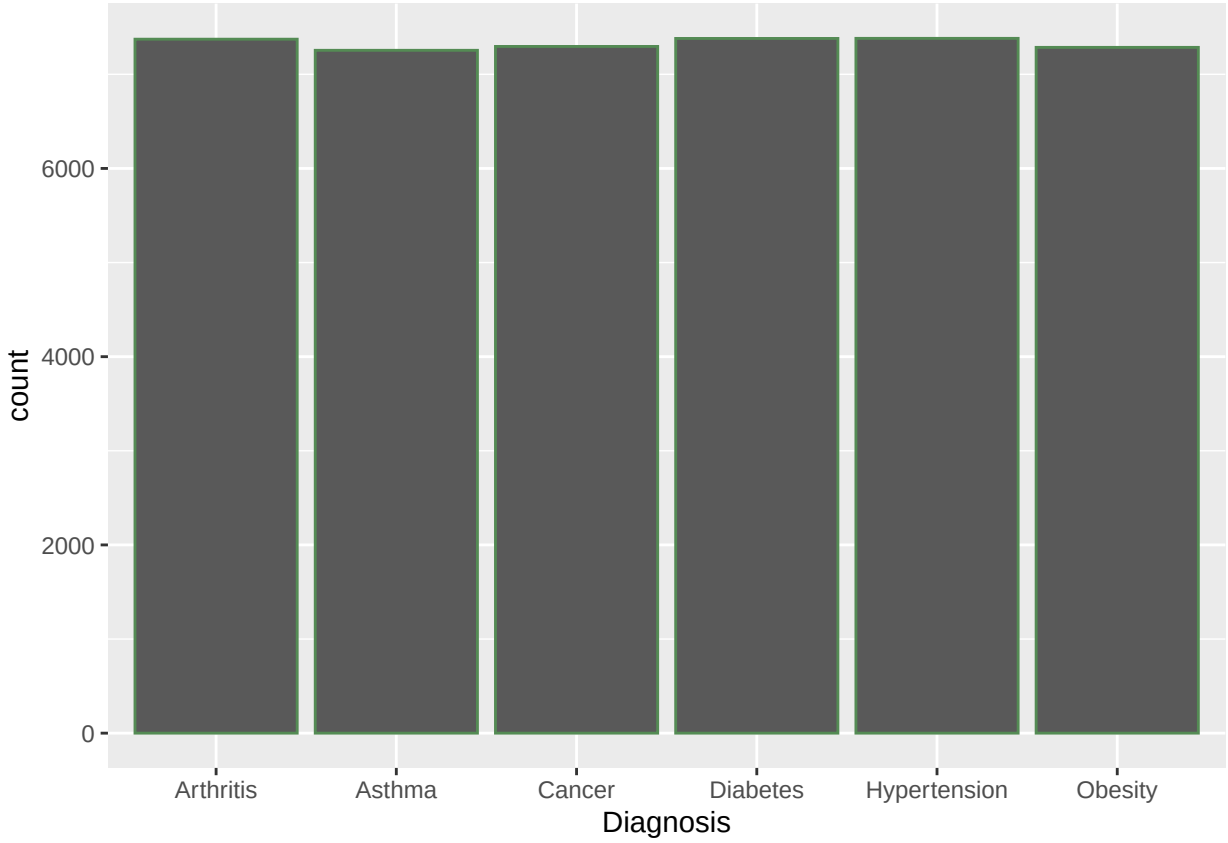
14. Plot a scatter plot for any 2 variables in the dataset.

```
ggplot(data = Training_healthcare, aes(x= Patient_Age, y= Medical_Bill)) + geom_point(color = "palegreen
```



15. Plot a bar plot for any 2 variables in the dataset.

```
ggplot(data = Training_healthcare, aes(x= Diagnosis)) + geom_bar(color = "palegreen4")
```



16. Find the correlation between any 2 variables in the dataset by applying Pearson correlation.

```
cor(Training_healthcare$Patient_Age,Training_healthcare$Medical_Bill,method = "pearson")
```

```
## [1] -0.003697453
```